

Introduction to Knot Theory

(9) Generalised Braid and Knot

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Recall: Braid Group B_n

Definition

The **Braid Group** B_n is defined by generators $\sigma_1, \dots, \sigma_{n-1}$ with:

$$B_n = \langle \sigma_1, \dots, \sigma_{n-1} \mid \mathcal{R} \rangle$$

where the relations $\mathcal{R}$ are:

$$\sigma_i \sigma_j = \sigma_j \sigma_i \quad \text{for } |i - j| \geq 2 \quad (1)$$

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \quad \text{for } 1 \leq i < n - 1 \quad (2)$$

- **Generators:** σ_i is the crossing of i -th and $(i + 1)$ -th strands.
- **Relations:** Braids are equivalent under ambient isotopy.

Recall: Alexander's and Markov's Theorem

Theorem (Alexander, 1923)

Every knot or link is isotopic to the closure of a braid.

Knot can be translated into braid, giving an algebraic representatives of knots.

Theorem (Markov, 1935)

Two braids have isotopic closures if and only if they are related by a finite sequence of Markov moves.

Recall: Markov's Theorem

Conjugation

$$\beta \sim \alpha\beta\alpha^{-1}$$

Sliding a braid around its closure does not change the knot type.

Stabilization

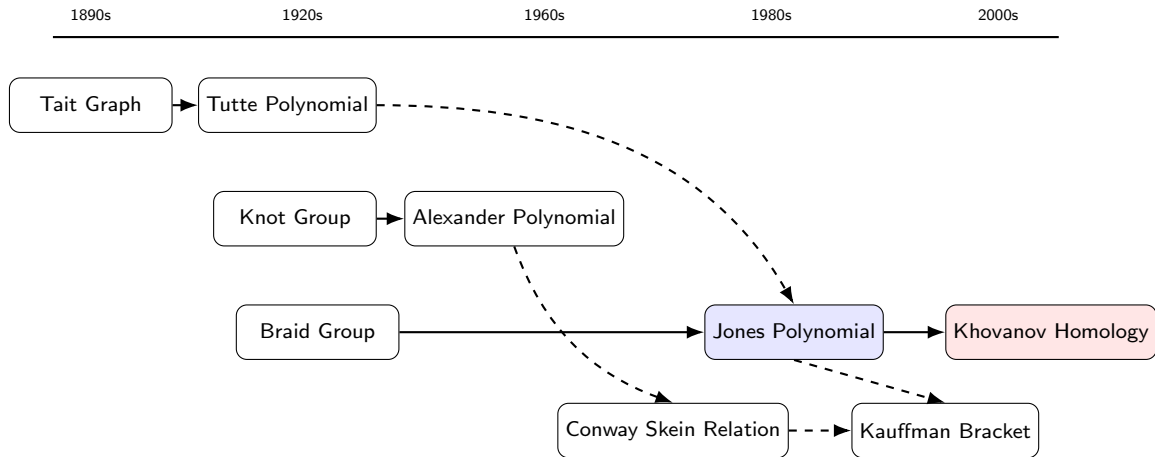
$$\beta \sim \beta\sigma_n^{\pm 1}$$

Adding a trivial strand preserves the closure.

The relation $\mathcal{R}$ comes from the isotopy of braid group itself.

The two relations $\mathcal{W}$ are only equivalent for the closure, so out of braid group.

History of Knot Invariants



Burau Representation

To construct a representation of B_n it suffices to find matrices satisfying the relation $\mathcal{R}$.

Definition

For a nonzero complex number t , a **Burau representation** is

$$\rho : B_n \rightarrow GL_n(\mathbb{C})$$

by sending σ_i to the matrix β_i , where β_i is defined as following.

$$\beta_i = \begin{pmatrix} I_{i-1} & & & \\ & 1-t & t & \\ & 1 & 0 & \\ & & & I_{n-i-1} \end{pmatrix}$$

A matrix with only nontrivial elements in rows and columns of $i, i+1$.

Burau Representation

The matrix β_i satisfies the relation $\mathcal{R}$. Hence the assignment

$$\sigma_i \mapsto \beta_i$$

defines a representation of the braid group B_n .

Reduced Burau representation

The subspace

$$V = \left\{ (x_1, \dots, x_n) \in \mathbb{C}^n \mid \sum_{i=1}^n x_i = 0 \right\}$$

is invariant under the Burau representation.

Taking the quotient by V yields the **reduced Burau representation** of dimension $n - 1$.

Alexander Polynomial

Theorem (Burau, 1936)

Let ψ denote the reduced Burau representation and let $\alpha \in B_n$.

Then

$$\det(I - \psi(\alpha))$$

is equal to

$$(1 + t + \cdots + t^{n-1})$$

times the Alexander polynomial of the closed braid $\hat{\alpha}$.

Intuition of Burau Representation

For a positive braid (not considering σ_i^{-1}), we can view Burau matrix as a stochastic process.

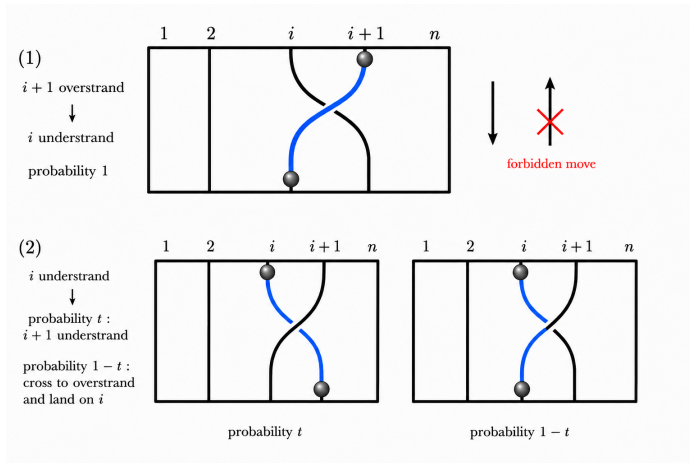


Figure: GPT-generated image

Intuition of Burau Representation

Consider braid group as a lift of a symmetric group (or a Coxeter group type A_{n-1}), by adding a relation $\sigma_i^2 = 1$.

Since β_i has $n - 2$ of eigenvalue 1 and only one eigenvalue of $-t$, so Burau representation is a deformation of a Coxeter reflection representation with parameter t on a Coxeter group type A_{n-1} .

The minimal polynomial for eigenvalues $1, -t$ is $(x - 1)(x + t) = 0$, so obtain a quadratic relation $(\sigma_i - 1)(\sigma_i + t) = 0$.

Jones Polynomial

Jones discovered the polynomial through von Neumann algebras, which later led to the Hecke algebra formulation.

Hecke Algebra

Define a Hecke algebra from group ring $\mathbb{C}[B_n]$ (or generally $A[B_n]$) by adding quadratic relation.

$$H(q, n) = \mathbb{C}[B_n] / \{(\sigma_i - q)(\sigma_i + 1)\}$$

For the given relation $(\sigma_i - q)(\sigma_i + 1) = 0$ i.e. $\sigma_i^2 = (q - 1)\sigma_i + q$, we can say every basis σ_i^k of B_n with $k \geq 2$ is linear combination of σ_i and 1.

Taking finite-dimensional deformation of braid, a good representation theory gives good trace.

Markov Trace

Markov Trace

A Markov trace is a family of traces $\text{tr} : H_n(q) \rightarrow \mathbb{C}$ satisfying

- $\text{tr}(ab) = \text{tr}(ba)$.
- $\text{tr}(1) = 1$.
- $\text{tr}(x\sigma_n) = z \cdot \text{tr}(x)$ for any $x \in H_n(q)$.

, where z is a complex number independent of q .

$H(q, n)$ is embedded in $H(q, n+1)$ by identifying σ_i s.

$$H(q, 1) \subset H(q, 2) \subset H(q, 3) \subset \cdots$$

so that it controls the stabilization condition of Markov theorem.

Jones Polynomial

For a braid $\beta \in B_n$, the closure $\hat{\beta}$ is a knot. Define Jones polynomial as

$$V_{\hat{\beta}}(q) = \alpha_n \cdot \text{tr}(\rho(\beta)),$$

where

$$\rho : B_n \rightarrow H_n(q)$$

is the natural projection to the Hecke algebra and α_n is a normalization factor depending on the writhe of β .

Jones Skein Relation

Using the quadratic relation, we get

$$\sigma_i - q\sigma_i^{-1} + (1 - q) = 0.$$

Let $\beta^+ = u\sigma_i v$ be braid, $\beta^- = u\sigma_i^{-1}v$ and $\beta^0 = uv$, where $u, v \in B_n$.

Taking the closure ($\hat{}$), we obtain skein triple.

$$K^+ = \hat{\beta}^+, \quad K^- = \hat{\beta}^-, \quad K^0 = \hat{\beta}^0$$

Taking the trace,

$$\begin{aligned} \text{tr}(\beta^+) - q \text{tr}(\beta^-) &= \text{tr}(u\sigma_i v) - q \text{tr}(u\sigma_i^{-1} v) \\ &= \text{tr}(uv(\sigma_i - q\sigma_i^{-1})) \\ &= (q - 1)\text{tr}(uv) = (q - 1)\text{tr}(\beta^0). \end{aligned}$$

Thus the Markov trace satisfies a skein-type relation.

$$\text{tr}(\beta^+) - q \text{tr}(\beta^-) = (q - 1)\text{tr}(\beta^0)$$

Redefine braid group

Definition (Pure braid group)

Let M be a manifold of dimension ≥ 2 , and $\prod_{i=1}^n M$ is n -fold product space.

Let $F_{0,n}M$ is a subspace

$$F_{0,n}M = \left\{ (z_1, \dots, z_n) \in \prod_{i=1}^n M \mid z_i \neq z_j \text{ if } i \neq j \right\}.$$

The fundamental group $\pi_1 F_{0,n}M$ of space $F_{0,n}M$ is the pure (or unpermuted) braid group with n strings of manifold M .

Definition (Full braid group)

Take an equivalence relation $z \sim z'$ for $z, z' \in F_{0,n}M$ if permutation of z_i s maps z to z' .

Let $B_{0,n}M$ is a quotient space $F_{0,n}M / \sim$.

The fundamental group $\pi_1 B_{0,n}M$ of space $B_{0,n}M$ is the full braid group of manifold M .

From Configuration Space to Geometric Braids

By taking M as Euclidean plane E^2 , $\pi_1 B_{0,n}E^2$ is a braid group B_n of Artin.

Choose a base point

$$z^0 = (z_1^0, \dots, z_n^0) \in F_{0,n}E^2$$

and let

$$p : F_{0,n}E^2 \rightarrow B_{0,n}E^2$$

be the quotient map.

Then an element of $\pi_1(B_{0,n}E^2, p(z^0))$ is represented by a loop

$$\ell : (I, \{0, 1\}) \rightarrow (B_{0,n}E^2, p(z^0)).$$

This loop lifts uniquely to a path $\tilde{\ell}$ with initial point $\tilde{\ell}(0) = z^0$ as

$$\tilde{\ell}(t) = (\ell_1(t), \dots, \ell_n(t)) \in F_{0,n}E^2.$$

Geometric Braid

Definition (Geometric Braid)

For the path $\tilde{\ell}$, which is a lift of a loop ℓ of $\pi_1(B_{0,n}E^2, p(z^0))$, each coordinate function defines an arc

$$Q_i = \{(\ell_i(t), t) \mid t \in I\} \subset E^2 \times I.$$

A **geometric braid** is a union of coordinate function as following.

$$Q = Q_1 \cup \cdots \cup Q_n$$

Geometric Braid

A geometric braid consists of n disjoint strands in

$$E^2 \times I$$

, because the points of $\tilde{\ell}(t) \in F_{0,n}E^2$ never collide, hence the strands remain disjoint.

The i -th braid string is the i -th strand

$$\mathcal{Q}_i = \{(\ell_i(t), t)\}.$$

Geometric braid is a representative of path class in the fundamental group $\pi_1 B_{0,n}E^2$.

Thus, $\pi_1(B_{0,n}E^2)$ gives unique braids up to homotopy relative to the base point.

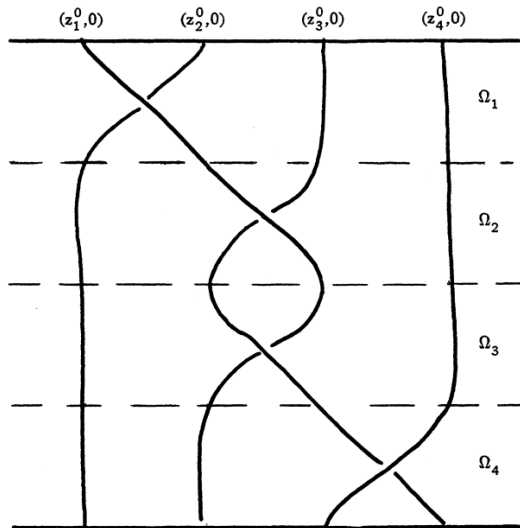


Figure: Geometric Braid

Equivalence of Braids

Two geometric braids Q, Q' are equivalent if the corresponding paths

$$\tilde{\ell}, \tilde{\ell}' : I \rightarrow F_{0,n}E^2$$

are homotopic relative to endpoints.

If $Q \sim Q'$, there exists

$$\mathcal{F} : I \times I \rightarrow F_{0,n}E^2$$

such that

$$\mathcal{F}(t, 0) = \tilde{\ell}(t), \quad \mathcal{F}(t, 1) = \tilde{\ell}'(t).$$

This produces a continuous deformation of geometric braids, and agrees with isotopy in $E^2 \times I$.

Equivalence of Braids

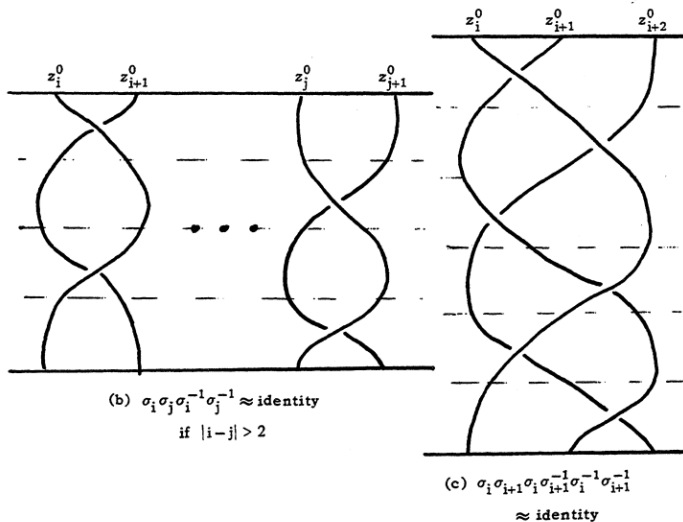


Figure: Equivalence relation of Braid Group

Equivalence of Braids

Theorem (Artin, 1925)

Two braids are equivalent if and only if there exists an isotopic deformation

$$G_s : E^2 \times I \rightarrow E^2 \times I$$

preserving the boundary planes $E^2 \times \{0\}$ and $E^2 \times \{1\}$.

Definition (Goldsmith's homotopy braid)

Allow each strand to pass through itself, but not through other strands.
Then one obtains the *homotopy braid group*

$$\hat{B}_n.$$

This is a quotient of the ordinary braid group:

$$B_n \twoheadrightarrow \hat{B}_n.$$

Equivalence of Braids

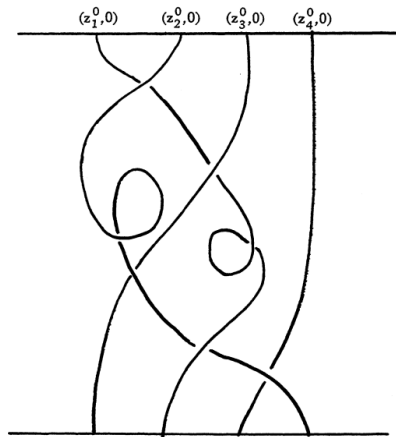


Figure: Intermediate state during the Goldsmith's homotopy

Motion Groups Perspective

Braids group can be viewed as a group of motions of a submanifold N in a manifold M .

Definition (Motion group)

A **motion** of N in M is a path

$$f_t \in \mathcal{H}(M)$$

with

$$f_0 = \text{id}_M, \quad f_1(N) = N.$$

Equivalence classes of motions form the **motion group** of N in M .

Two motions f and g are equivalent if the path $f^{-1}g$ is homotopic modulo its endpoint to a stationary motion.

In this perspective, $\pi_1(M)$ is group of motions of an interior point in M .

Configuration Spaces

Covering Projection

The quotient map

$$p : F_{0,n}M \rightarrow B_{0,n}M$$

is a regular covering map.

The (deck) transformation group Σ_n is group of all permutations of n letters, so full symmetric group. There is a canonical isomorphism

$$\pi_1 B_{0,n}M / \pi_1 F_{0,n}M \cong \Sigma_n.$$

Fadell–Neuwirth Fibration

Theorem (Fadell–Neuwirth, 1962)

Let a forgetful map $\pi : F_{m,n}M \rightarrow F_{m,r}M$ for $1 \leq r < n$ be

$$\pi(z_1, \dots, z_n) = (z_1, \dots, z_r).$$

Then π exhibits $F_{m,n}M$ as locally trivial fiber space over the base space $F_{m,r}M$ with fiber $F_{m+r,n-r}M$.

Geometrically, π is a map that fix first r particles and let the remaining $n - r$ particles move freely.

Fadell–Neuwirth Fibration

Proof.

For $F_{m,r}M$ take finite m points Q_m .

The fiber $\pi^{-1}(z_1^0, \dots, z_r^0)$ be

$$\{(z_1^0, \dots, z_r^0, y_{r+1}, \dots, y_n) \mid z_1^0, \dots, z_r^0, y_{r+1}, \dots, y_n \text{ are distinct and in } M - Q_m\}$$

Selecting $Q_{m+r} = Q_m \cup \{z_1^0, \dots, z_r^0\}$,

$$F_{m+r,n-r}M = \{(y_{r+1}, \dots, y_n) \mid y_{r+1}, \dots, y_n \text{ are distinct and in } M - Q_{m+r}\}$$



Inductive Structure of Configuration Spaces

Repeatedly applying the forgetful fibration gives

$$F_{m,n}M \rightarrow F_{m,n-1}M \rightarrow \cdots \rightarrow F_{m,1}M.$$

Each step adds one moving particle in a punctured manifold.

The fiber of

$$F_{m,n}M \rightarrow F_{m,n-1}M$$

is

$$M - Q_{m+n-1}.$$

Thus configuration spaces are constructed recursively by adding particles one at a time.

Example: Configuration Space of the Plane

Take $M = E^2$, and map $\pi : F_{0,n}E^2 \rightarrow F_{0,n-1}E^2$.

Fixing $(z_1, \dots, z_{n-1})$, the remaining point moves in

$$E^2 - \{z_1, \dots, z_{n-1}\}.$$

Hence the fiber is

$$E^2 - Q_{n-1}.$$

The free group naturally appears inside braid groups as following.

$$\pi_1(E^2 - Q_{n-1}) \cong \text{Free}(n-1)$$

Since $\pi_2(E^2 - Q_m) = \pi_3(E^2 - Q_m) = \pi_0(M - Q_m) = 1$ for every $m \geq 0$, induce an exact sequence of homotopy groups,

$$1 \longrightarrow \pi_1(E^2 - Q_{n-1}) \longrightarrow \pi_1(F_{0,n}E^2) \longrightarrow \pi_1(F_{0,n-1}E^2) \longrightarrow 1.$$

Thus pure braid groups are constructed inductively by repeatedly adding one moving particle (adding one strand).

Braids on General Manifolds

The topology of the manifold M strongly affects the braid group

$$B_n(M) = \pi_1(B_{0,n}M).$$

Theorem (Birman, 1969)

Let M be any compact surface except

$$S^2 \quad \text{or} \quad \mathbb{RP}^2.$$

Then the classical braid group injects into the surface braid group:

$$B_n(E^2) \hookrightarrow B_n(M).$$

Similarly,

$$P_n(E^2) \hookrightarrow P_n(M).$$

Exceptional Surfaces

The sphere and projective plane behave differently.

Exceptional Surfaces

$$S^2, \quad \mathbb{RP}^2$$

do not contain the classical braid group faithfully.

For example, spherical braid groups satisfy additional relations not in Artin braid groups.

Example

In $B_n(S^2)$, one has an extra relation of the form

$$\sigma_1 \cdots \sigma_{n-1}^2 \cdots \sigma_1 = 1.$$

Generalised Braid

Key Principle

Braids are fundamental groups of configuration spaces.

Change the ambient space:

$$\mathbb{R}^2 \rightsquigarrow \Sigma$$

Change the moving objects:

points $\rightsquigarrow$ circles, ribbons, surfaces.

New braid groups arise automatically.

Loop Braid

Definition (Configuration space of circles)

Let R_n be the space of configurations of n disjoint, unlinked Euclidean circles in E^3 .

$$R_n = \{(C_1, \dots, C_n) \mid C_i \cap C_j = \emptyset\}.$$

The circles are allowed to move in 3-space, but they must remain disjoint, unlinked, and unknotted.

Definition (Loop braid group)

Let

$$C = C_1 \sqcup \dots \sqcup C_n$$

be a trivial link of n circles in E^3 .

The loop braid group is

$$LB_n \cong \pi_1(R_n),$$

describing motions of circles in E^3 up to isotopy.

From Braids to Loop Braids

Circles trace annuli in

$$E^3 \times I.$$

Thus loop braids are naturally related to *ribbon* surfaces in 4-space.

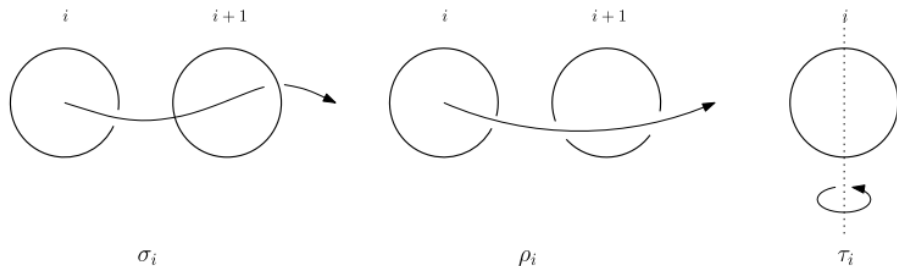


Figure: Motion of circles in E^3

Ribbon Braid

As particles trace strands, circles trace annuli.

$$S^1 \times I$$

Hence loop braids can be represented by embedded annuli in

$$E^3 \times I \cong E^4.$$

These are called ribbon braids.

Theorem (Damiani, 2017)

The ribbon braid group rB_n is isomorphic to the loop braid group:

$$rB_n \cong LB_n.$$

Virtual Knots as Knots in Surfaces

A classical knot is an embedding $S^1 \hookrightarrow S^3$ or equivalently a knot diagram in the plane modulo Reidemeister moves.

Virtual Knots

Virtual knot theory extends classical knot theory by allowing knots to live in

$$\Sigma_g \times [0, 1],$$

where Σ_g is an orientable surface of genus g .

A configuration space of Σ_g is called a surface braid group.

Virtual braids arise by considering braid motions on surfaces together with stabilization and destabilization of the underlying surface.

Virtual Braid Group

Definition (Virtual braid group)

The virtual braid group VB_n is generated by

$$\sigma_1, \dots, \sigma_{n-1}, \quad v_1, \dots, v_{n-1},$$

where:

- σ_i are classical braid generators,
- v_i are virtual crossings.

The relations consist of braid relations for σ_i , symmetric group relations for v_i , with $\sigma_i v_j = v_j \sigma_i$ for $|i - j| \geq 2$ and mixed relations $v_i v_{i+1} \sigma_i = \sigma_{i+1} v_i v_{i+1}$.

$$VB_n \cong B_n *_{S_n} S_n.$$

Virtual Knots

Definition (Virtual Crossing)

A **virtual crossing** is neither an overcrossing nor an undercrossing. It represents an artifact of projecting a knot diagram on a surface to the plane.

- Classical crossings satisfy the usual Reidemeister moves.
- Virtual crossings are marked by a small circle.
- Additional equivalence: detour move allows a strand containing only virtual crossings to be freely moved.

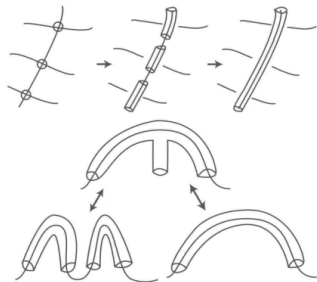


Figure: Detour move

Alexander and Markov Theorems for Virtual Knots

Theorem (Kamada, 2007)

Every virtual knot or link can be represented as the closure of a virtual braid.

Thus, virtual knots also admit algebraic representatives through the virtual braid group VB_n .

Theorem (Kamada, 2007)

Two virtual braids have equivalent virtual link closures if and only if they are related by a finite sequence of virtual Markov moves.

Consequently, the study of virtual knots can be reduced to the study of virtual braid groups and their equivalence relations.

Virtual Reidemeister Moves

Generalized Reidemeister Moves

Virtual knot theory is generated by Classical Reidemeister moves, Virtual Reidemeister moves, and Mixed (semivirtual) moves.

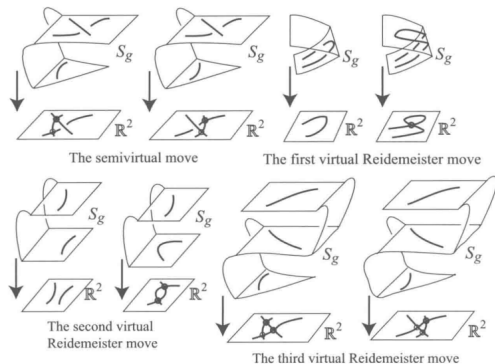


Figure: Generalized Reidemeister moves and thickened surfaces

Virtual Knots and Thickened Surfaces

Theorem (Carter–Kamada–Saito, 2002; Kuperberg, 2003)

Virtual links are equivalent to stable equivalence classes of links in thickened surfaces.

Let Σ be a compact oriented surface and L a link diagram on Σ
 (Σ_1, L_1) and (Σ_2, L_2) are equivalent if there exists a compact orientable surface Σ_3 and orientation preserving map $f_i : \Sigma_i \rightarrow \Sigma_3$ s.t. $f_i(L_i)$ are related by classical Reidemeister moves.

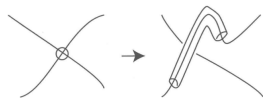


Figure: Lift virtual crossing to handle

Forbidden Moves

Forbidden moves

Certain moves are not allowed in virtual knot theory. Admitting both forbidden moves trivializes the theory.

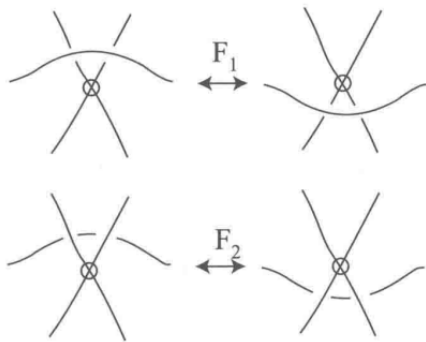


Figure: Forbidden moves

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